Informe del Compilador HULK

David Sánchez

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Abstract

Contents

References

[1] Aho, A. V., Lam, M. S., Sethi, R., Ullman, J. D. (2007). *Compilers: Principles, Techniques, and Tools*. Pearson/Addison Wesley.

A Gramática Completa

```
input -> line
    | lines_block
    | line lines_block

lines_block -> { lines }

lines -> epsilon
    | non_empty_lines

non_empty_lines -> line
    | non_empty_lines line

line -> expr ;
    | func_assign ;
    | type_node_decl

expr -> arit_op
    | bool_expr
    | while while_expr
    | string
    | id_expr
    | func_call
    | let_assign
    | if conditional
    | member_access_expr
    | expr := expr
    | expr = expr
    | expr @ expr
    | expr @@ expr
    | expr as var_id

func_assign -> function var_id ( args_list ) => expr
    | function var_id ( args_list ) : type_id => expr
    | function var_id ( args_list ) { lines }
    | function var_id ( args_list ) : type_id { lines }

args_list -> epsilon
    | id_expr
    | args_list , id_expr

id_expr -> var_id : type_id
    | var_id

let_assign -> let var_assign_list in expr
    | let var_assign_list in lines_block

var_assign_list -> id_expr = expr
    | id_expr = expr as var_id
    | id_expr = new_expr
    | var_assign_list , id_expr = expr
    | var_assign_list , id_expr = expr as var_id
    | var_assign_list , id_expr = new_expr

new_expr -> new var_id ( expr_list )

expr_list -> epsilon
```

```
    | expr
    | new_expr
    | expr_list , expr
    | expr_list , new_expr

func_call -> var_id ( expr_list )

arit_op -> number
    | ( expr )
    | expr + expr
    | expr - expr
    | expr * expr
    | expr / expr
    | expr ^ expr
    | expr % expr
    | - expr

bool_expr -> bool
    | expr >= expr
    | expr > expr
    | expr <= expr
    | expr < expr
    | expr == expr
    | expr != expr
    | expr & expr
    | expr | expr
    | ! expr
    | id_expr is type_id
    | func_call is var_id

conditional -> ( bool_expr ) expr else expr
    | ( bool_expr ) lines_block else expr
    | ( bool_expr ) expr else lines_block
    | ( bool_expr ) lines_block else lines_block
    | ( bool_expr ) expr elif conditional
    | ( bool_expr ) lines_block elif conditional

while_expr -> ( bool_expr ) lines_block
    | ( bool_expr ) expr

type_node_decl -> type type_id { type_body_elements }
    | type type_id inherits type_id { type_body_elements }
    | type type_id inherits type_id ( args_list ) { type_body_elements }
    | type type_id ( args_list ) { type_body_elements }
    | type type_id ( args_list ) inherits type_id { type_body_elements }
    | type type_id ( args_list ) inherits type_id ( args_list ) { type_body_elements }

type_body_elements -> epsilon
    | type_body_elements attribute
    | type_body_elements method

attribute -> id_expr = expr ;
    | id_expr = expr as var_id ;

method -> var_id ( args_list ) => expr ;
    | var_id ( args_list ) : type_id => expr ;
    | var_id ( args_list ) { lines }
    | var_id ( args_list ) : type_id { lines }
```

B Fragmentos de Código Fuente

Listing 1: Ejemplo de código C++

```
// Tu c digo aqu
```